



University of Debrecen, Faculty of Informatics

Logic in computer science

First-order logic

(predicate logic)

central semantic notions

November 18, 2022

First-order model

Let $\Gamma \subset \text{Form}$ be a nonempty set of formulae of a first-order language $\mathcal{L}^{(1)}$, $\langle U, \varrho \rangle$ be an interpretation of the language and ν be an assignment relying on $\langle U, \varrho \rangle$.

The ordered triple $\langle U, \varrho, \nu \rangle$ is a *model* of Γ if $|A|_{\nu}^{\langle U, \varrho \rangle} = 1$ for all $A \in \Gamma$.

Satisfiability (SAT)

Let $\Gamma \subseteq \text{Form}$ be an arbitrary set of formulae of an $\mathcal{L}^{(1)}$ first-order language.

- The Γ set of formulae is *satisfiable*, if it has a model.
- The Γ set of formulae is *unsatisfiable*, if it is not satisfiable.

Let $A \in \text{Form}$ be an arbitrary formula of an $\mathcal{L}^{(1)}$ first-order language.

- The formula A is *satisfiable* if the $\{A\}$ singleton set is satisfiable.
- The formula A is *unsatisfiable* if the $\{A\}$ singleton set is unsatisfiable.

Semantic equivalence

Let $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first-order language, $\Gamma \subseteq Form$ be an arbitrary set of formulae, and $A \in Form$ be an arbitrary formula of the language.

- The formula A is the *semantic consequence* of Γ if $\Gamma \cup \{\neg A\}$ is unsatisfiable.

(Notation: $\Gamma \models A$)

- The formula A is *valid* (*logical law*) if $\emptyset \models A$.

(Notation: $\models A$)

Logical equivalence

Let $\mathcal{L}^{(1)} = \langle \mathcal{LC}, \mathcal{Var}, \mathcal{Con}, \mathcal{Term}, \mathcal{Form} \rangle$ be a first-order language and $A \in \mathcal{Form}$ and $B \in \mathcal{Form}$ be two formulas of $\mathcal{L}^{(0)}$.

The formula A and formula B are *logically equivalent*, if

$$A \models B \quad \text{and} \quad B \models A.$$

(Notation: $A \Leftrightarrow B$)



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standard and prenex form

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Variable replacement and renaming

Free variable replacement

Let $A \in \text{Form}$ be a formula, $x \in \text{Var}$ and $y \in \text{Var}$ be variables. The $[A_y^x]$ denotes the formula where all free occurrences of x is replaced by y .

Bound variable renaming

Renaming the bound variable x to y in the formula QxA has the result: $Qy[A_y^x]$.

The renaming is correct if

- $y \notin \text{FreeVar}(A)$, and
- there is no free occurrence of x in A which is in the scope of a quantifier bounding the variable y .

A renaming is correct, if y does not occur in the formula.

Standard form

A formula A is in *standard form* if

- the free and bound variables are different ($FreeVar(A) \cap BounVar(A) = \emptyset$),
- there is no two different quantifier bounding the same variable.

Theorem

Let A be a formula of a first-order language $\mathcal{L}^{(1)}$. There exists a formula in standard form which is semantically equivalent with A .

Prenex form

Let $\langle \mathcal{L}, \text{Var}, \text{Con}, \text{Term}, \text{Form} \rangle$ be a first-order language. The formula

$$Q_1 x_1 Q_2 x_2 \dots Q_n x_n A$$

is in *prenex form* if A is a quantification free formula (A contains no quantifiers) and

- $Q_i \in \{\forall, \exists\}$ and,
- $x_i \in \text{Var}$

for all $i \in \{1, \dots, n\}$ where $n \geq 0$.

Theorem

Let A be a formula of a first-order language $\mathcal{L}^{(1)}$. There exists a formula in prenex form which is semantically equivalent with A .

Remark

The goal of the prenex form is to maximize the scope of the quantifiers.